



Mathematics 2 (5EZB0) Lecture 4

Koondi Mitra

*Mathematics & Computer Science
Department*



Limit

Limit

$$\lim_{x \rightarrow x_0} f(x) = L$$

1. If x_0 is not an isolated point, i.e., for all $\delta > 0$
 $B_\delta(x_0) \cap \mathcal{D} \neq \{x_0\}$

2. if for every $\epsilon > 0$,
there exists $\delta > 0$, such that
 $|f(x) - L| < \epsilon$ for all $x \in B_\delta(x_0) \cap \mathcal{D}$

➤ The limit, if existing, is unique

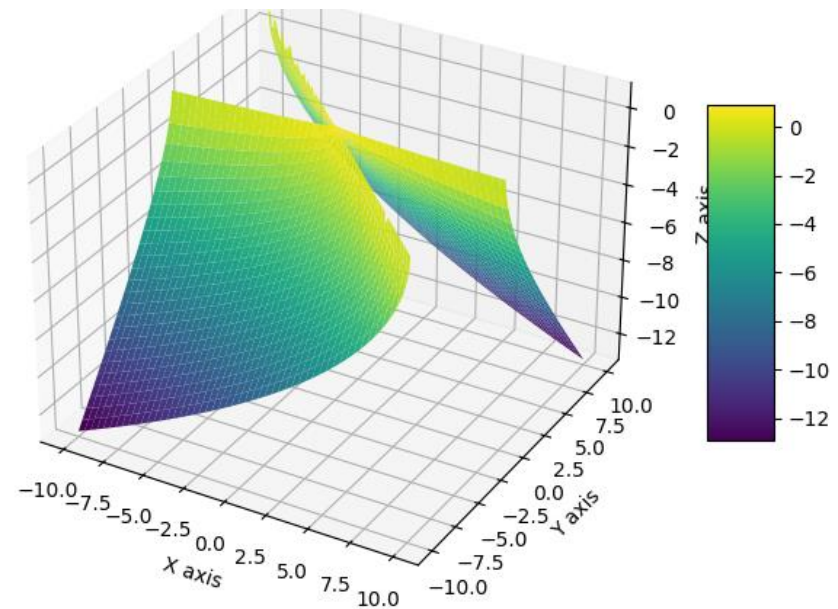
Continuous functions

A function f is continuous at $x = x_0$ if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

$$f(x, y) = 1 - \sqrt{xy + y^2}$$

1. What is the maximum domain of this function?



2. What is its limit at (0,0)?

Partial derivatives

Let the function f be continuous at a point
 $\mathbf{x} = (x_1, x_2, \dots, x_d) \in \mathcal{D}$.

Its partial derivatives are then defined as

$$\frac{\partial f}{\partial x_1}(\mathbf{x}) = \lim_{\Delta x_1 \rightarrow 0} \frac{f(x_1 + \Delta x_1, x_2, \dots, x_d) - f(x_1, x_2, \dots, x_d)}{\Delta x_1}$$

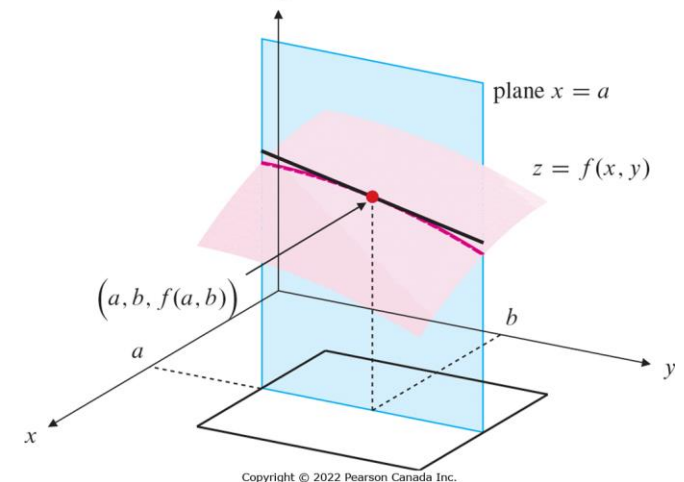
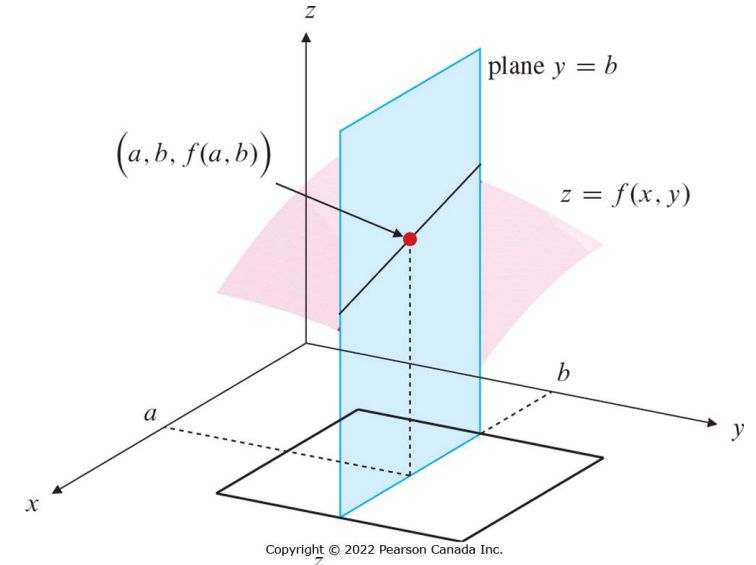
$$\frac{\partial f}{\partial x_2}(\mathbf{x}) = \lim_{\Delta x_2 \rightarrow 0} \frac{f(x_1, x_2 + \Delta x_2, \dots, x_d) - f(x_1, x_2, \dots, x_d)}{\Delta x_2}$$

$\vdots$

If these limits exist.

$$f(x, y) = 1 - \sqrt{xy + y^2}$$

1. Find $\partial_x f$.
2. Is it continuous at $(0,0)$?



Second order derivatives

Let the function f have continuous partial derivatives at a point $\mathbf{x} = (x_1, x_2, \dots, x_d) \in \mathcal{D}$. Then its higher order derivatives are defined as

$$\frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) = \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) (\mathbf{x}) = \lim_{\Delta x_1 \rightarrow 0} \frac{\partial_{x_1} f(x_1 + \Delta x_1, x_2, \dots, x_d) - \partial_{x_1} f(x_1, x_2, \dots, x_d)}{\Delta x_1}$$

$$\frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) = \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) (\mathbf{x}) = \lim_{\Delta x_2 \rightarrow 0} \frac{\partial_{x_1} f(x_1, x_2 + \Delta x_2, \dots, x_d) - \partial_{x_1} f(x_1, x_2, \dots, x_d)}{\Delta x_2}$$

If these limits exist.

➤ $\frac{\partial^2 f}{\partial x_2 \partial x_1}$ is also denoted as $\partial_{x_2 x_1} f$, $f_{x_2 x_1}$, and f_{21} .

Question 1.1

Show that
 $w(x, t) = f(x - ct) + g(x + ct)$,

satisfies the *Wave equation*:

$$\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2}.$$

Question 1.2

Find all the second order partial derivatives of
 $f(x, y, z) = e^{3x+4y} \sin(5z)$.

Show that f satisfies the *Laplace equation*:

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0.$$

Second order derivatives

Let the function f have continuous partial derivatives at a point $\mathbf{x} = (x_1, x_2, \dots, x_d) \in \mathcal{D}$.
Then its higher order derivatives are defined as

$$\frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) = \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) (\mathbf{x}) = \lim_{\Delta x_1 \rightarrow 0} \frac{\partial_{x_1} f(x_1 + \Delta x_1, x_2, \dots, x_d) - \partial_{x_1} f(x_1, x_2, \dots, x_d)}{\Delta x_1}$$

$$\frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) = \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) (\mathbf{x}) = \lim_{\Delta x_2 \rightarrow 0} \frac{\partial_{x_1} f(x_1, x_2 + \Delta x_2, \dots, x_d) - \partial_{x_1} f(x_1, x_2, \dots, x_d)}{\Delta x_2}$$

If these limits exist.

Schwarz/Clairaut theorem

Let $f: \mathcal{D} \rightarrow \mathbb{R}$ have continuous second order partial derivatives at a point $\mathbf{x}_0 = (x_1, x_2, \dots, x_d) \in \mathcal{D}$.
Then for all $1 \leq i, j \leq d$,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}_0)$$

For an example where it does not hold, see [wiki/Symmetry_of_second_derivatives](https://en.wikipedia.org/wiki/Symmetry_of_second_derivatives)

The chain rule

Let the two functions $x(t, \cdot), y(t, \cdot)$ depend on t .

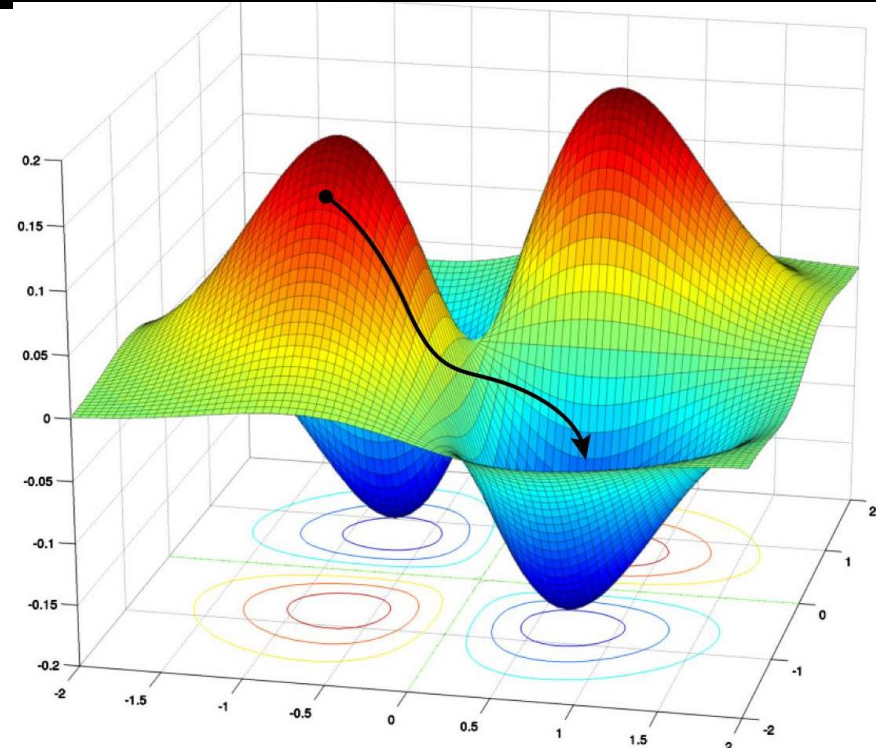
Consider a function $f: \mathcal{D} \rightarrow \mathbb{R}$.

How does $f(x(t, \cdot), y(t, \cdot))$ change with t ?

Chain rule

If f, x, y are all continuously differentiable, then

$$\frac{\partial}{\partial t} (f(x(t, \cdot), y(t, \cdot))) = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}$$



source: sciencesprings.wordpress.com

Question 2.1

Take the mountain profile $H(x, y) = \sin(x) \sin(y)$. You are hiking (descending) along the path $y = x^2 - 2$ with speed $\sqrt{5}$ m/s in the xy -plane. What is your velocity at $(-1, -1)$?

The chain rule

Let the two functions $x(t, \cdot), y(t, \cdot)$ depend on t .

Consider a function $f: \mathcal{D} \rightarrow \mathbb{R}$.

How does $f(x(t, \cdot), y(t, \cdot))$ change with t ?

Chain rule

If f, x, y are all continuously differentiable, then

$$\frac{\partial}{\partial t} (f(x(t, \cdot), y(t, \cdot))) = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}$$

Question 2.2

The temperature profile of a region is

$$T(x, y, z, t) = \frac{xy}{1+z} (1+t).$$

A balloon rises and flows along the wind as

$$x(t) = t, \quad z(t) = 1 + t^2.$$

What is the rate of change of the temperature?

Question 2.3

1. If $f(x, y)$ is *harmonic* (satisfies the Laplace equation), then show that $g(x, y) = f(x^2 - y^2, 2xy)$ is also harmonic.

Tangent planes and normals

- The tangent plane of a surface $z = f(x, y)$ at a point (x_0, y_0) is

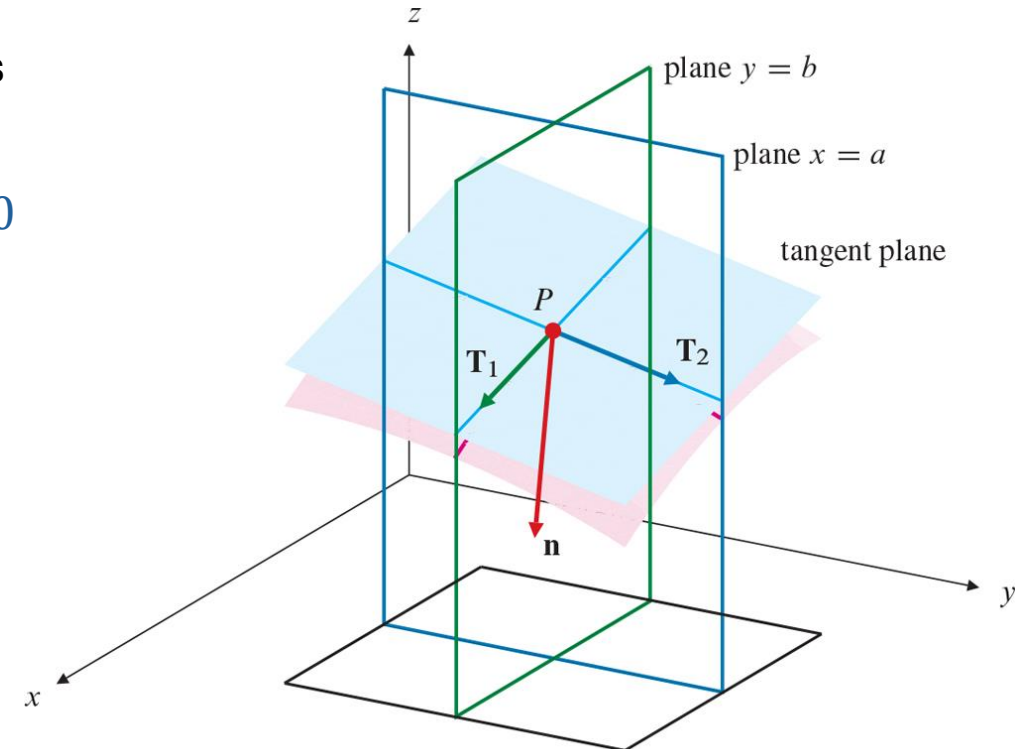
$$\frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) - (z - f(x_0, y_0)) = 0$$

- The normal to this plane at point (x_0, y_0) is

$$\left(\frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0), -1 \right)$$

- The normal line is hence

$$\frac{x - x_0}{\frac{\partial f}{\partial x}(x_0, y_0)} = \frac{y - y_0}{\frac{\partial f}{\partial y}(x_0, y_0)} = \frac{z - f(x_0, y_0)}{-1}$$



Copyright © 2022 Pearson Canada Inc.

Question 3

Take the function $z = f(x, y) = 4 - x^2 + 2xy - 2y^2$.

1. What is the tangent plane at $(0, 1)$?
2. At which point is the tangent plane horizontal?
3. Can you spot something special about this point?